

Multiple myeloma & precursors

MGUS → smoldering → myeloma · **treat on a myeloma-defining event, not on the M-spike** · anti-CD38 quadruplets up front, T-cell redirection (CAR-T, bispecifics) moving to first relapse

Compiled from primary trial reports and consensus guidelines (IMWG; NCCN) verified against PubMed, September 2026.

THE PLASMA CELL SPECTRUM (IMWG 2014)

Entity	Definition
MGUS <i>~3% of adults >50</i>	M-protein <3 g/dL + clonal marrow PCs <10% + no CRAB / SLiM (or amyloidosis). Progresses ~1%/yr; IgM MGUS → WM, not myeloma
Light-chain MGUS	Abnormal FLC ratio + raised involved FLC, no heavy chain on IFE , PCs <10%, urine M <500 mg/24 h
Smoldering (SMM)	M-protein ≥3 g/dL (or urine ≥500 mg/24 h) and/or PCs 10–60% , no CRAB / SLiM. ~10%/yr for the first 5 yr, then ~3%/yr, then ~1%/yr
Multiple myeloma	Clonal PCs ≥10% (or biopsy-proven plasmacytoma) + ≥1 myeloma-defining event (below)
Plasma cell leukemia	≥5% circulating PCs (IMWG 2021; was 20%) — treat as ultra-high risk
Solitary plasmacytoma	Single lesion, marrow <10% clonal (none = solitary; minimal = higher risk). RT 40–50 Gy

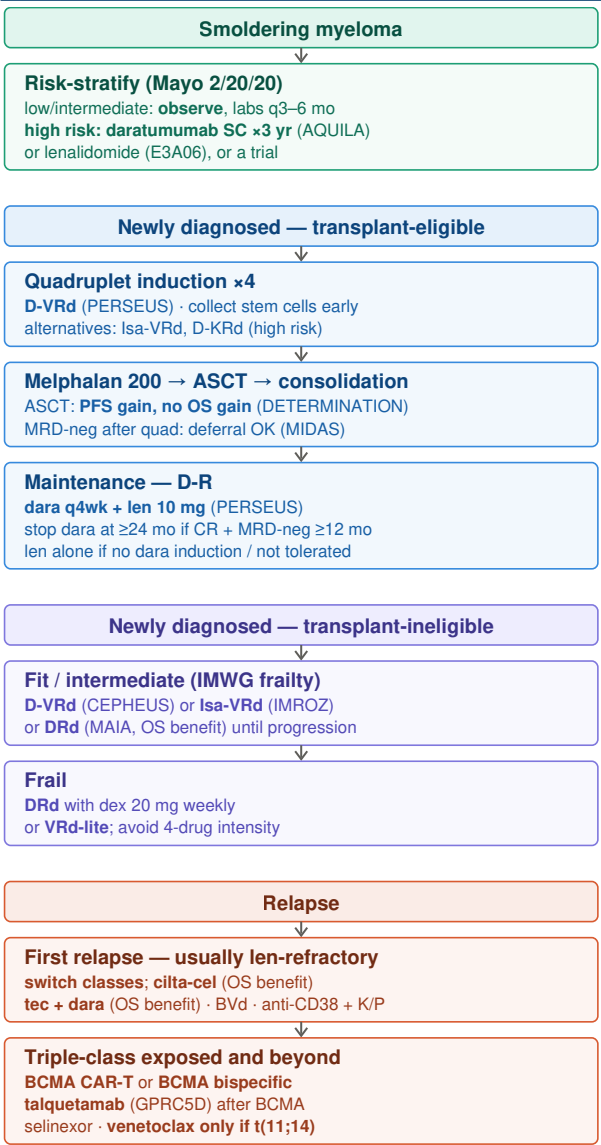
MYELOMA-DEFINING EVENTS — ANY ONE = TREAT

- **C** — Ca >11 mg/dL or >1 mg/dL above ULN · **R** — CrCl <40 or Cr >2 mg/dL · **A** — Hb <10 g/dL or >2 below LLN · **B** — ≥1 **lytic lesion** (≥5 mm) on X-ray, CT or PET-CT
- **SLiM** (no CRAB needed): clonal PCs ≥60% · involved:uninvolved **FLC ratio ≥100** with involved FLC ≥100 mg/L · >1 **focal lesion on MRI** (≥5 mm)
- Osteoporosis / compression fracture **without a lytic lesion** is not CRAB — image first

WORKUP

- **SPEP + serum IFE + serum FLC** — plus 24-h UPEP + urine IFE once at diagnosis
- CBC, CMP (Ca, Cr, albumin), **β2-microglobulin**, LDH, quantitative IgG/IgA/IgM
- **Marrow aspirate + biopsy**: PC %, flow, **FISH on CD138-sorted cells** — t(4;14), t(14;16), t(14;20), t(11;14), del(17p), 1q+, del(1p); **TP53** sequencing where available
- **Whole-body low-dose CT** or **PET-CT**; **whole-body MRI** if SMM with a negative CT/PET. **Skeletal survey is obsolete** (insensitive)
- Albuminuria or unexplained organ dysfunction → think **AL amyloidosis** (fat pad, NT-proBNP, troponin)
- **Type & screen with extended RBC phenotype/genotype before the first anti-CD38 dose** (page 3)

TREATMENT PATHWAY



SMOLDERING MYELOMA — WHO TO TREAT

- **Mayo 2/20/20**: M-protein >2 g/dL, marrow PCs >20%, FLC ratio >20. 0 / 1 / ≥2 factors → median time to progression **110 / 68 / 29 months**
- **High-risk SMM** → **daratumumab SC ×3 yr** (AQUILA vs active monitoring): PFS HR **0.49**, 5-yr PFS 63% vs 41%, **OS HR 0.52** — **FDA 6 Nov 2025**, the first approved SMM therapy
- **Lenalidomide** (E3A06, intermediate/high risk): PFS HR 0.28; no OS signal
- **Low-risk SMM: observe** — treating it is not supported

MGUS — MAYO RISK (20-YR PROGRESSION)

Risk factors	20-yr risk
M-protein ≥1.5 g/dL · non-IgG · abnormal FLC ratio — 0 factors	5%
1 · 2 · 3 factors	21% · 37% · 58%

Low-risk MGUS (IgG, M <1.5, normal ratio): **no marrow or imaging needed**; recheck at 6 mo then every 2–3 yr. Anything else: marrow + imaging, then yearly labs.

STAGING

System	Definition · outcome
ISS	I: β2M <3.5 mg/L + albumin ≥3.5 g/dL · III: β2M ≥5.5 · II: neither
R-ISS (2015)	I: ISS I + standard-risk FISH + normal LDH · III: ISS III + high-risk FISH [del(17p), t(4;14), t(14;16)] or high LDH · II: rest. 5-yr OS 82 / 62 / 40%
R2-ISS (2022)	Points: ISS III 1.5, ISS II 1, del(17p) 1, high LDH 1, t(4;14) 1, 1q+ 0.5 → I 0 · II 0.5–1 · III 1.5–2.5 · IV 3–5. Median OS NR / 109 / 69 / 38 mo. Splits the big R-ISS II group

HIGH-RISK MYELOMA — IMS/IMWG 2025 (ANY ONE)

- **del(17p)** in >20% cancer clonal fraction and/or TP53 mutation
- **IgH translocation** [t(4;14), t(14;16), t(14;20)] + **1q gain/amp** and/or **del(1p32)** — translocation **alone no longer qualifies**
- **Biallelic del(1p32)**, or **monoallelic del(1p32) + 1q gain/amp**
- **β2M ≥5.5 mg/L with normal creatinine** (<1.2 mg/dL)
- Also high risk in practice: **plasma cell leukemia**, extramedullary disease, **relapse <12–18 mo** from front-line therapy (functional). **t(11;14), t(6;14), hyperdiploidy = standard risk**

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NEWLY DIAGNOSED — THE TRIALS THAT SET THE STANDARD

Trial	Design · result
PERSEUS <i>TE</i>	D-VRd vs VRd induction/consolidation + ASCT, then D-R vs R : 6-yr PFS 81.1% vs 56.1% (HR 0.35, 81-mo update, meeting report); MRD-neg 75% vs 48% . Dara stopped after ≥24 mo if sustained MRD-neg. FDA 30 Jul 2024 for D-VRd induction + consolidation only (not D-R maintenance)
DETERMINATION · IFM 2009	RvD ± ASCT, then len: PFS 67.5 vs 46.2 mo (IFM: 50 vs 36) — no OS difference (5-yr 80.7% vs 79.2%). ASCT buys PFS, not OS ; deferral with cells banked is legitimate
MIDAS <i>IFM 2020-02</i>	Isa-KRd ×6, then MRD-guided. MRD-neg (10^{−5}): ASCT vs 6 more Isa-KRd — MRD-neg 10 ^{−6} 86% vs 84% (P=0.64). MRD+: tandem vs single ASCT 32% vs 40% (NS). MRD endpoint only , 17-mo follow-up, no PFS yet
Len maintenance <i>meta-analysis</i>	After ASCT: PFS 52.8 vs 23.5 mo , OS HR 0.75 ; more second primary malignancies
AURIGA	MRD+ after ASCT: D-R vs R maintenance — MRD-neg conversion at 12 mo 50.5% vs 18.8% ; PFS HR 0.53
MAIA <i>TIE</i>	DRd vs Rd until progression: median PFS NR vs 34.4 mo (HR 0.53), OS HR 0.68
SWOG S0777	VRd vs Rd : PFS 43 vs 30 mo, OS 75 vs 64 mo (HR 0.71)
CEPHEUS <i>TIE / deferred</i>	D-VRd vs VRd ×8 → D-Rd vs Rd: MRD-neg 60.9% vs 39.4% ; PFS HR 0.57 . FDA 27 Jan 2026 (TIE only)
IMROZ <i>TIE, ≤80 yr</i>	Isa-VRd vs VRd : 5-yr PFS 63.2% vs 45.2% (HR 0.60). FDA Sep 2024

No head-to-head quadruplet vs DRd in TIE. Quadruplets for fit older adults; DRd for intermediate-fit / frail.

FITNESS AND DOSE ADJUSTMENT

- **IMWG frailty score** (age, Charlson, ADL, IADL; 0–5): **fit 0 · intermediate 1 · frail ≥2** — 3-yr OS 84% / 76% / 57%; frail have ~2× the discontinuation rate
- Age >75 or frail: **dexamethasone 20 mg weekly, weekly SC bortezomib**, renally dosed lenalidomide

RESPONSE (IMWG 2016)

Level	Criteria
MRD-neg	CR + no clonal PCs at ≤10 ^{−5} (NGF or NGS); sustained = confirmed ≥1 yr apart; imaging MRD-neg adds PET-negative
sCR / CR	Negative serum + urine IFE , plasmacytomas gone, <5% marrow PCs (sCR adds normal FLC ratio, no clonal PCs by IHC)
VGPR / PR	≥90% M-protein fall (or IFE+ / SPEP−) / ≥50% serum and ≥90% urine fall
Progression	≥25% rise from nadir: serum M ≥ 0.5 g/dL absolute , urine ≥200 mg/24 h, or dFLC >10 mg/dL; new lesions

MRD is prognostic, and now regulatory: ODAC 12–0 (Apr 2024) accepted MRD-neg CR as an accelerated-approval endpoint. CEPHEUS (Jan 2026) used MRD-neg as its primary endpoint; **iberdomide (Aug 2026)** = **first accelerated approval of a new drug based on MRD**. Stopping therapy on MRD alone is still a trial question.

RELAPSE — BY WHAT THE PATIENT IS REFRACTORY TO

Setting	Options · evidence
Definitions	Triple-class exposed = PI + IMiD + anti-CD38. Refractory = progression on or ≤60 days after. Penta = 2 PIs + 2 IMiDs + anti-CD38. Treat biochemical relapse early if high-risk or rapid rise
Len-refractory, 1–3 lines <i>the usual first relapse</i>	Cilta-cel (CARTITUDE-4): PFS NR vs 11.8 mo (HR 0.26), OS HR 0.55 . Teclistamab + dara (MajesTEC-3): 36-mo PFS 83% vs 30% (HR 0.17), OS HR 0.46 . BVd (DREAMM-7): PFS 36.6 vs 13.4 mo, OS HR 0.58
Anti-CD38-naïve, 1–3 lines	Dara-Kd (CANDOR, HR 0.63) · Isa-Kd (IKEMA, HR 0.53) · DPd (APOLLO, 12.4 vs 6.9 mo) · Isa-Pd (ICARIA, 11.5 vs 6.5 mo) · IberDd (iberdomide, Zenbexus ; accelerated Aug 2026, ≥1 prior line incl PI + IMiD)
Anti-CD38 + len-refractory	Teclistamab monotherapy (MajesTEC-9 vs PVd/Kd): PFS HR 0.29 , OS HR 0.60 , grade 3–4 infection 42%. BPd (DREAMM-8 vs PVd): 12-mo PFS 71% vs 51%. CAR-T
≥4 lines, triple-class	BCMA CAR-T, BCMA or GPRC5D bispecific (table below) · SVd (BOSTON: PFS 13.9 vs 9.5 mo) · clinical trial

Cilta-cel vs ide-cel: ide-cel (KarMMa-3, 2–4 prior lines) PFS 13.3 vs 4.4 mo but **no OS gain** (HR 1.01, 56% crossover). **Both CAR-Ts show early excess deaths** during manufacturing/bridging — bridge effectively.

T-CELL REDIRECTION AND ADC — APPROVALS

Agent	Target · label · key data
Cilta-cel (CAR-T)	BCMA · ≥1 prior line, len-refractory · OS benefit . Delayed parkinsonism , cranial-nerve palsy (9%), GBS
Ide-cel (CAR-T)	BCMA · ≥2 prior lines (IMiD, PI, anti-CD38) · PFS benefit, OS not shown
Teclistamab	BCMA×CD3 SC · + dara, ≥1 line (5 Mar 2026) · mono ≥4 lines (ORR 63%) converted to regular approval 5 Mar 2026
Elranatamab	BCMA×CD3 SC · ≥4 lines (accelerated; ORR 61%)
Linvoseltamab	BCMA×CD3 IV · ≥4 lines (accelerated Jul 2025; ORR 71%, ≥CR 50%)
Talquetamab	GPRC5D ×CD3 SC · ≥4 lines (accelerated; ORR ~70%) · works after BCMA ; oral/skin/nail toxicity, fewer infections
Belantamab (ADC)	BCMA · BVd, ≥2 lines incl PI + IMiD (re-approved Oct 2025; withdrawn 2022 after DREAMM-3) · keratopathy , REMS

Sequencing: BCMA loss after BCMA therapy is common — after BCMA CAR-T/bispecific, **switch target (GPRC5D)** rather than re-using BCMA when possible.

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REGIMEN DOSING

Regimen	Dosing
VRd 21-day	Bortezomib 1.3 mg/m² SC d1, 4, 8, 11 + lenalidomide 25 mg d1–14 + dex 20 mg d1, 2, 4, 5, 8, 9, 11, 12 (S0777, x8)
D-VRd 28-day (PERSEUS)	Dara SC 1800 mg weekly c1–2, q2wk c3–6 + V 1.3 d1, 4, 8, 11 + R 25 d1–21 + dex 40 d1–4, 9–12 x4 → ASCT → x2 → D-R (dara q4wk)
VRd-lite 35-day	Bortezomib 1.3 SC weekly d1, 8, 15, 22 + len 15 d1–21 + dex 20 twice weekly
DRd 28-day (MAIA)	Dara SC 1800 mg: weekly wk 1–8, q2wk wk 9–24, q4wk from wk 25 + len 25 d1–21 + dex 40 weekly (20 if >75)
Isa-VRd (IMROZ)	Isatuximab 10 mg/kg IV weekly x4, then q2wk; later q4wk + VRd x4 → Isa-Rd
KRd 28-day	Carfilzomib 20 → 27 mg/m² d1, 2, 8, 9, 15, 16 (or 56–70 once weekly) + len 25 d1–21 + dex 40 weekly
Dara-Kd / Isa-Kd	Carfilzomib 20 → 56 mg/m² d1, 2, 8, 9, 15, 16 + dex 40 weekly + anti-CD38
DPd / Isa-Pd	Pomalidomide 4 mg d1–21 q28d + dex 40 weekly (20 if ≥75) + dara SC or isatuximab 10 mg/kg
BVd	Belantamab 2.5 mg/kg IV q3wk + V 1.3 d1, 4, 8, 11 + dex 20 x8 cycles, then belantamab alone. Eye-toxicity reductions: 1.9 mg/kg q3wk → 1.9 q8wk
SVd	Selinexor 100 mg PO once weekly + bortezomib 1.3 SC weekly + dex 20 twice weekly (35-day)
IberDd (Zenbexus)	Iberdomide 1 mg d1–21 q28d + dara SC + dex 40 weekly
Dara SC (AQUILA, SMM)	1800 mg weekly c1–2, q2wk c3–6, then q4wk — stop at 36 mo (39 cycles)
Len maintenance	10 mg d1–28 (15 after 3 mo if tolerated); CrCl 30–60 → 5 mg; <30 → 2.5 mg; dialysis 5 mg after HD
Mel 200 + ASCT	Melphalan 200 mg/m² day –2; 140 mg/m² if renal impairment or frail (100–140 on dialysis); oral cryotherapy
Teclistamab SC	0.06 → 0.3 mg/kg step-up, then 1.5 mg/kg weekly; q2wk after ≥CR ≥6 mo. With dara: 1.5 weekly wk 2–8 → 3 mg/kg q2wk wk 9–24 → q4wk
Eliranatamab SC	12 mg d1 → 32 mg d4 → 76 mg weekly; q2wk from wk 25 if responding, q4wk from wk 49
Talquetamab SC	0.01 → 0.06 mg/kg → 0.4 mg/kg weekly, or + 0.4 step → 0.8 mg/kg q2wk
Linvoseltamab IV	5 → 25 → 200 mg (d1, 8, 15), 200 weekly wk 4–13, q2wk from wk 14; q4wk from wk 24 if ≥VGPR
CAR-T	Flu 30 + Cy 300 mg/m² x3 d. Cilta-cel 0.5–1.0x10 ⁶ CAR+ cells/kg · ide-cel 300–510x10 ⁶ . Stay near center 2 wk, no driving 2 wk (REMS removed Jun 2025)

Lenalidomide renal dosing (induction): CrCl 30–60 → 10 mg daily · <30 → 15 mg q48h · dialysis → 5 mg daily after HD. Bortezomib, daratumumab, carfilzomib, pomalidomide (off HD) need no renal adjustment; bortezomib: reduce for hepatic impairment.

SUPPORTIVE CARE

- **Bone:** zoledronic acid 4 mg IV q4wk (q12wk after ~1–2 yr) or denosumab 120 mg SC q4wk for every patient on active therapy (non-inferior, HR 0.98; less renal toxicity 10% vs 17%, more hypocalcemia 17% vs 12%). Denosumab if CrCl <30; don't stop it abruptly (rebound fractures). Zoledronic acid renal dosing: CrCl 50–60 → 3.5 mg · 40–49 → 3.3 · 30–39 → 3.0 · <30 → avoid
- **ONJ:** dental evaluation before starting; avoid extractions on therapy. Calcium + vitamin D for all (not if hypercalcemic)
- **VTE with any IMiD:** aspirin 81–325 mg if ≤1 risk factor; prophylactic LMWH or apixaban 2.5 mg BID / rivaroxaban 10 mg if higher risk (IMPEDE VTE ≥4, prior VTE, high-dose dex, doxorubicin, immobility)
- **Infection:** acyclovir/valacyclovir with Pls, anti-CD38, CAR-T, bispecifics. Levofloxacin 500 mg x12 wk at diagnosis (TEAMM: fever or death HR 0.66). PJP prophylaxis with bispecifics, CAR-T, high-dose dex
- **Bispecifics / CAR-T:** IVIG 0.4 g/kg q4wk if IgG <400 mg/dL (hypogammaglobulinemia is near-universal with BCMA agents). Vaccinate: inactivated flu, pneumococcal conjugate, COVID, RSV, recombinant zoster; no live vaccines; re-vaccinate after ASCT

MYELOMA EMERGENCIES

- **Renal failure / cast nephropathy** (very high involved FLC): start a bortezomib-based regimen immediately (e.g. D-VCd / VRd with renal dosing), high-dose dex, hydrate, treat Ca; stop NSAIDs, avoid IV contrast. Plasma exchange and high-cutoff dialysis are not routine (negative trials). Biopsy if albuminuria → AL, MIDD
- **Hypercalcemia:** IV saline, zoledronic acid (or denosumab if renal failure), calcitonin for the first 48 h, steroids + anti-myeloma therapy
- **Cord compression:** dexamethasone 10 mg IV now + MRI whole spine → RT or surgery (instability) · Hyperviscosity (IgA, IgG3): plasmapheresis · Anemia: treat the myeloma; ESA if Hb <10 despite therapy (VTE risk with IMiDs)

PITFALLS WITH MODERN AGENTS

- **Anti-CD38** → **blood bank:** dara/isa bind CD38 on reagent RBCs → **panreactive indirect antiglobulin test** for up to 6 mo. Send **type & screen + phenotype/genotype before dose 1**; lab uses **DTT-treated cells** (DTT destroys Kell → give **K-negative units**). ABO typing is unaffected — tell the blood bank
- **Anti-CD38** → **SPEP/IFE:** dara (IgGκ) shows up as a small **IgG-κ band** that mimics residual disease — use a DIRA / mass-spec assay before calling non-CR
- **CRS / ICANS:** step-up dosing, **48-h admission** (or monitored outpatient) after step-ups; tocilizumab for grade ≥2 CRS, dex for ICANS; **restart step-up if doses are missed**
- **Cilta-cel:** **delayed movement/neurocognitive toxicity (parkinsonism)** weeks–months out, cranial-nerve (VII) palsy; **boxed warning for secondary T-cell malignancies** (all CAR-T, Apr 2024), MDS/AML
- **Talquetamab:** **dysgeusia, dry mouth, dysphagia, weight loss, skin peeling, nail loss** — common and dose-limiting; dose spacing helps
- **Belantamab:** **corneal keratopathy** (ocular AEs 79–89% in DREAMM-7/8) — blurred vision, dry eye; **eye exam at baseline and before doses**; hold/delay by exam findings; REMS

OTHER AGENTS — WHEN AND WHEN NOT

- **Venetoclax:** **t(11;14) only, off-label.** BELLINI (Vd ± ven): PFS gain **but excess deaths** (infections), driven by non-t(11;14) — **never outside t(11;14)**
- **Selinexor** (XPO1): SVd weekly — thrombocytopenia 39% grade 3–4, nausea, **hyponatremia**, fatigue; give with 5-HT3 + olanzapine prophylaxis
- **Carfilzomib:** HTN, HF, cardiac events — **baseline echo and BP control**; prefer bortezomib if significant cardiac disease